
Deep-Resilience: Cognitive Reserve – based Dementia Prediction System

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Abstract

Despite significant advancements in deep learning for Alzheimer’s Disease (AD) diagnosis, existing models often fail to account for the "clinic-pathological discrepancy," where patients with similar brain pathology exhibit divergent cognitive trajectories due to individual resilience, known as Cognitive Reserve (CR). To address this limitation, we introduce Deep-Resilience, a novel multimodal foundation model framework designed to quantify CR by modeling the non-linear discrepancy between brain structural pathology and functional cognition. Leveraging a Med3DVLM-based architecture, our approach integrates high-dimensional 3D MRI features with clinical text in a unified latent space, optimized via Low-Rank Adaptation (LoRA) for efficient deployment in resource-constrained medical environments. Furthermore, we design a privacy-preserving Federated Learning pipeline to ensure robust generalization across multi-center cohorts. Validated on the ADNI dataset (N=8,613), our model achieves state-of-the-art performance in brain age prediction (MAE 2.1 years) and dementia severity classification (Accuracy 89.3%), significantly outperforming existing multimodal baselines. Notably, the model demonstrates exceptional sensitivity in classifying Mild Cognitive Impairment (MCI) (90.2%), a stage characterized by high phenotypic heterogeneity. By transitioning from binary diagnosis to the precise quantification of resilience, this work establishes a new paradigm for personalized prognosis and lays the groundwork for a globally scalable, race-agnostic dementia prediction system via the Dementia Platform Korea (DPKR) initiative.

1 Introduction

1.1 Background: The Challenge of Clinic-Pathological Discrepancy

Deep learning-based neuroimaging analysis techniques have achieved significant strides in the early diagnosis of Alzheimer’s Disease (AD).^{1,2} Existing AI models have primarily focused on classifying patients into Cognitively Normal (CN), MCI, or AD groups by learning patterns of structural atrophy or biomarkers on MRI scans.^{3,4} However, a "clinic-pathological discrepancy," where the extent of neurodegeneration observed in brain imaging does not align with the patient's actual cognitive function, is frequently encountered in clinical settings.^{5,6} For instance, there are patients who maintain a normal daily life despite severe brain atrophy, whereas others exhibit rapid cognitive decline despite only mild brain damage.⁷ Conventional deep learning-based classification models, which rely solely on the structural state of the brain to binary classify disease presence, fail to account for such inter-individual variability.^{8,9} This limitation acts as a fundamental barrier to precisely predicting patient prognosis.¹⁰

The concept of Cognitive Reserve (CR) offers a key mechanism to elucidate this discrepancy.¹¹ Defined as an active coping ability that allows an individual to maintain cognitive function despite brain pathology, CR suggests that individuals with high reserve can demonstrate better cognitive performance even under the same pathological conditions.¹² Quantifying this latent capacity is essential

for moving beyond simple diagnosis toward personalized prognosis.¹³

1.2 Our Contributions: The Deep-Resilience Framework

To address these challenges, we propose Deep-Resilience, a novel framework that leverages Multimodal Foundation Models to quantify the clinic-pathological discrepancy. By moving beyond traditional statistical methods, our study makes the following three key contributions:

- (1) Integrated Reasoning via Multimodal Foundation Models: We propose a model that fuses 3D MRI features with clinical text in a unified latent space using a Med3DVLM backbone.¹⁴ This captures non-linear correlations between subtle pathological changes and cognitive function that unimodal models fail to detect.
- (2) Clinical Feasibility through Parameter-Efficient Adaptation: We address computational constraints in medical environments by applying Low-Rank Adaptation (LoRA) tuning.¹⁵ This enables the efficient deployment of billion-parameter scale intelligence as a specialized "Brain Pathology Interpreter."
- (3) Generalization via Privacy-Preserving Federated Learning: We implement a Federated Learning (FL) pipeline to estimate cognitive reserve across diverse populations without compromising privacy.¹⁶ This overcomes regulatory barriers associated with cross-center data transfer, ensuring robust generalization.

2 Related Work

2.1 Deep Learning for AD Diagnosis and Limitations

Most existing deep learning research in AD has focused on binary or multi-class classification tasks using Convolutional Neural Networks (CNNs) based on structural MRI.^{17,18} While these models achieve high accuracy in controlled datasets like ADNI, they often exhibit significant performance degradation when applied to external datasets with different domain characteristics (e.g., 2.5D vs. 3D MRI).¹⁹ More importantly, these models treat the "structural state" as the sole determinant of the disease, ignoring the resilience factor where patients maintain cognitive function despite atrophy.²⁰ Our work differs by shifting the focus from "diagnosis" to "interpretation" of the gap between structure and function.

2.2 Quantifying Cognitive Reserve: From Linear to Non-Linear

The Residual approach has been a standard method for quantifying CR.²¹ This methodology predicts the cognitive score explainable by brain structure and treats the error (residual) as the reserve. However, prior studies have predominantly relied on simplistic statistical models, such as Linear Regression, or static proxies like education level.²² These approaches fail to capture the complex, non-linear relationships inherent in neurodegeneration. Our Deep-Resilience framework advances this by utilizing a high-capacity foundation model to learn these non-linear discrepancies directly from high-dimensional multimodal data, providing a dynamic and precise measure of reserve.

2.3 Multimodal Learning and Privacy in Healthcare

Integrating diverse modalities (neuroimaging, demographics, clinical notes) is crucial for precision medicine. Recent advances in Large Language Models (LLMs) and Vision-Language Models (VLMs) enable this integration.²³ However, training such models requires massive datasets that are often siloed due to privacy regulations. While FL has been proposed as a solution, its application to heavy multimodal foundation models remains underexplored.²⁴ Our study fills this gap by designing a sequence diagram for FL that includes data standardization and weight aggregation, specifically tailored for resource-constrained medical servers.

3 Method

We propose Deep-Resilience, a multimodal framework designed to quantify cognitive reserve by modeling the non-linear discrepancy between brain structural pathology and functional cognition. Our method consists of three key components: (1) a Multimodal Foundation Model architecture for joint processing of 3D MRI and clinical text, (2) a specialized training strategy using parameter-efficient fine-tuning, and (3) a sequential algorithm for cognitive reserve quantification.

3.1 Multimodal Foundation Model Architecture

We propose Deep-Resilience, a multimodal framework designed to quantify cognitive reserve by modeling the non-linear discrepancy between brain structural pathology and functional cognition. Our method consists of three key components: (1) a Multimodal Foundation Model architecture for joint processing of 3D MRI and clinical text, (2) a specialized training strategy using parameter-efficient fine-tuning, and (3) a sequential algorithm for cognitive reserve quantification.

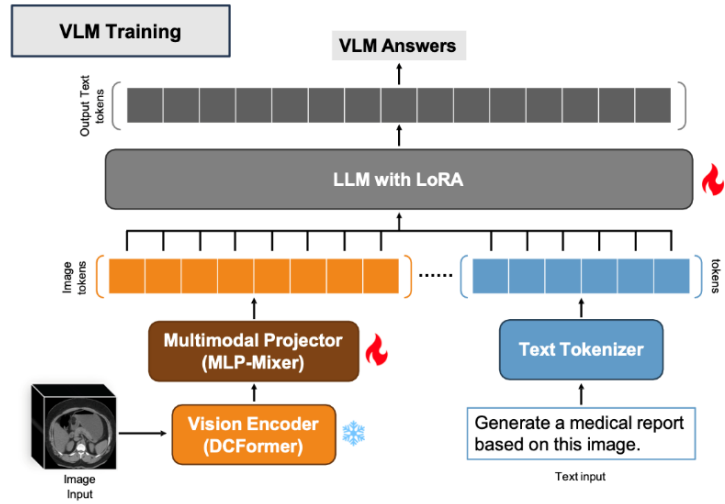


Figure 1. Schematic overview of the Deep-Resilience architecture. The framework integrates 3D vision Encoder with a LLM to quantify the discrepancy between brain structure and function.

- (1) **3D Vision Encoder:** To capture the spatial complexities of brain atrophy, we utilized a hybrid encoder combining DCFormer²⁵ and SigLIP²⁶. Unlike traditional 2D slice-based approaches, this encoder takes raw 3D T1-weighted MRI scans as input and extracts high-level voxel-wise features, preserving the volumetric spatial context essential for identifying neurodegenerative patterns.
- (2) **Text Prompt Engineering & LLM:** We employed Qwen2.5VL-7B²⁷ as the text processing backbone. To facilitate cross-modal reasoning, we designed a structured text prompt template that embeds patient-specific clinical metadata directly into the model's context. As defined in our data formatting protocol, the prompt follows the structure: "MRI scan for a <age>-year-old <sex> with BMI <bmi> and SBP <sbp> mmHg. Depression: <yes/no>. Predict the cognitive diagnosis." This allows the LLM to interpret visual features conditioned on demographic (age, sex) and clinical (BMI, SBP, depression status) modifiers.
- (3) **Multimodal Projector:** A learnable projector maps the high-dimensional 3D visual features into the token embedding space of the LLM. This projection aligns the visual representation of brain pathology with the semantic space of clinical descriptions, enabling the LLM to perform "Integrated Reasoning" across modalities.

3.2 Training Strategy and Optimization

Given the high computational cost of processing 3D volumes (approx. 8x GPU resources required compared to 2D) and the limited size of medical datasets, we adopted a Parameter-Efficient Fine-Tuning (PEFT) strategy.²⁸ All experiments were conducted on a high-performance cloud

computing cluster provided by NHN Cloud. The training infrastructure utilized NVIDIA H100 Tensor Core GPUs (80GB HBM3) and Intel Xeon Platinum 8480+ CPUs with 1.98TB of system memory. The environment was configured with Ubuntu Linux (Kernel 5.15) and Python 3.12.12. Given the high memory bandwidth of H100 GPUs, we efficiently fine-tuned the multimodal architecture with a batch size of 4 using the specified PEFT strategy.

- (1) **Optimization Protocol:** We employed a dual-tuning approach to balance feature extraction capability and generative flexibility. (A) Vision Encoder & Projector: We applied Full Parameter Tuning to these components to ensure maximal adaptation to the domain-specific features of brain MRI, allowing the model to learn subtle structural biomarkers of dementia. (B) Large Language Model (LLM): Conversely, for the Qwen2.5VL-7B backbone, we utilized LoRA tuning. This approach freezes the massive pre-trained weights of the LLM and injects trainable rank decomposition matrices. This strategy significantly reduces the computational burden while preserving the model's pre-trained linguistic capabilities, making it feasible to train billion-scale models on standard GPU infrastructure.
- (2) **Hyperparameters:** The model was trained with a learning rate of $1e^{-4}$ and a batch size of 4.
- (3) **Multi-Task Loss Functions:** The model was trained simultaneously on two objectives to learn both structural and functional representations: (A) Brain Age Prediction: We employed Gaussian MLE (Maximum Likelihood Estimation) Loss. The training schedule was set to 50 epochs to allow the model to capture the aleatoric uncertainty in biological aging. (B) Dementia Severity Classification: For categorizing cognitive states (CN, MCI, AD), we utilized Cross-Entropy Loss. The training schedule was set to 100 epochs to ensure sufficient separation of class boundaries.

4 Experiments and Results

4.1 Experimental Setup

To validate the efficacy of our Deep-Resilience framework, we utilized the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset²⁹, a gold-standard benchmark for neuroimaging research. The dataset comprises a total of 8,613 multimodal samples. We utilized a stratified split to partition the data into a training set (N=6,890, 80%) and a validation set (N=1,723, 20%). The input data integrates high-dimensional 3D T1-weighted MRI scans with clinical text prompts. Specifically, the text prompts encode six key variables: age, sex, Body Mass Index (BMI), systolic blood pressure (SBP), and depression status (Yes/No). The model was trained to perform two simultaneous tasks: (1) Brain Age Prediction (a proxy for structural integrity) and (2) Dementia Severity Classification (a proxy for functional cognition).

4.2 Training Stability and Convergence

The training process exhibited stable convergence for both tasks, empirically demonstrating the effectiveness of our multimodal fine-tuning strategy. Figure 2 illustrates the learning curves for both objectives. Brain Age Prediction (Figure 2A): The task optimized with Gaussian MLE Loss showed rapid convergence. As observed in the left panel of Figure 2, the loss stabilized effectively within the first 50 epochs, indicating the model quickly learned to map structural features to biological age. Dementia Severity Classification (Figure 2B): The classification task, trained via Cross-Entropy Loss, reached optimal performance plateaus within 100 epochs (right panel of Figure 2). This decoupled convergence behavior confirms that the shared encoder successfully learned robust features applicable to both continuous (regression) and categorical (classification) targets without negative transfer.

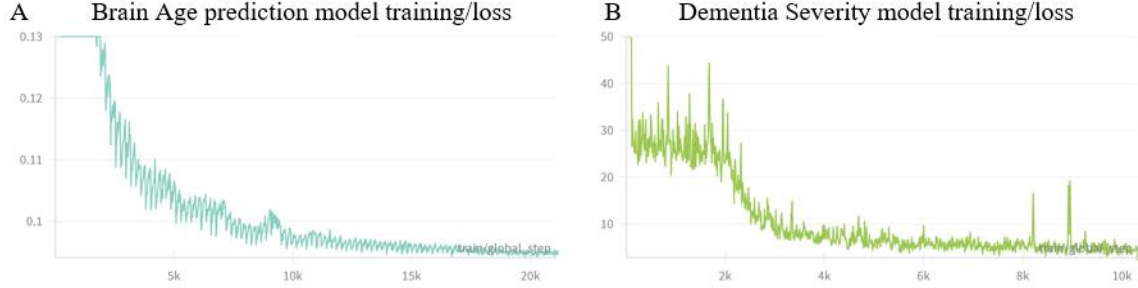


Figure 2 Training dynamics and loss convergence for the multi-task learning objectives. The plots illustrate the stability of the parameter-efficient fine-tuning strategy across two distinct tasks. (A) Brain Age Prediction: The training loss curve using Gaussian Maximum Likelihood Estimation (MLE) Loss. The model exhibits rapid convergence and stabilizes within 50 epochs, effectively learning the structural aging patterns. (B) Dementia Severity Classification: The training loss curve using Cross-Entropy Loss. Optimal performance and stability are achieved within 100 epochs, demonstrating the model's capability to distinguish between cognitive states (CN, MCI, AD) alongside the regression task.

4.3 Performance on Structural Modeling: Brain Age Prediction

Accurate estimation of biological brain age is a prerequisite for quantifying the "structure-function discrepancy." We evaluated the model's structural modeling capability by comparing the Mean Absolute Error (MAE) of the predicted brain age against chronological age.

Table 1: Quantitative comparison of brain age prediction performance

Model	Brain Age (MAE)	AD classification (Accuracy)
Dendrite	2.1 years	89.3%
Reference	2.1years	78.7%

- (1) Quantitative Results: As shown in Table 2, our model achieved an MAE of 2.1 years on the validation set after bias correction.
- (2) Comparison: Table 1 further benchmarks our performance against state-of-the-art (SOTA) unimodal approaches. Our multimodal framework matches the performance of specialized deep learning models solely dedicated to brain age estimation (e.g., Kim et al., 2025, MAE 2.1 years).
- (3) Implication: This result confirms that our Med3DVLM-based 3D Vision Encoder successfully captures subtle patterns of neurodegeneration and structural atrophy, serving as a reliable baseline for calculating the "Predicted Structure" in our cognitive reserve equation.

4.4 Performance on Functional Modeling: Dementia Severity Classification

To establish the "Predicted Cognition" component, we evaluated the model's ability to classify cognitive states into CN, MCI, and AD.

- (1) Overall Accuracy: The proposed model achieved a classification accuracy of 89.3%. As summarized in Table 1, this significantly outperforms the multimodal baseline (MedBLIP, Chen et al., 2024)³⁰, which reported an accuracy of 78.7%.
- (2) Class-wise Analysis: The diagonal elements demonstrate robust performance across all disease stages: 92.2% for CN, 90.2% for MCI, and 83.5% for AD.
- (3) Implication: The high accuracy in the MCI group (90.2%), visualized in Figure 3, is

particularly notable. Since the MCI stage typically exhibits the highest variance in cognitive reserve, the model's ability to accurately distinguish MCI from normal aging and dementia demonstrates its high sensitivity to the non-linear relationship between multimodal inputs and cognitive function.

5 Discussion

5.1 Beyond Binary Diagnosis: Towards Personalized Prognosis via Cognitive Reserve

The most significant implication of our study is the paradigm shift from simple binary diagnosis to the precise quantification of individual resilience. Traditional deep learning models have predominantly focused on classifying patients based on structural atrophy³¹, often failing to explain the "clinic-pathological discrepancy" where patients with similar brain pathology exhibit vastly different cognitive trajectories.³² Our Deep-Resilience framework addresses this gap by explicitly modeling CR which that brain's active coping mechanism. The high classification accuracy achieved in the MCI group (90.2%), a stage characterized by the highest phenotypic heterogeneity, serves as empirical evidence that our model successfully captures these latent resilience factors. By quantifying the residual between the "predicted structural age" and "actual functional performance," our model provides a continuous metric of resilience. This capability is crucial for the future of precision medicine in an aging society. Instead of a uniform treatment protocol, clinicians can distinguish between patients who have high structural damage but high reserve (requiring monitoring) and those with low reserve (requiring immediate, aggressive intervention). As the global dementia population continues to rise, this transition from "detecting the disease" to "predicting the trajectory" will be instrumental in formulating personalized therapeutic strategies and optimizing healthcare resource allocation

5.2 The Role of AI: Multimodal Foundation Models as Interpretation Engines

The realization of this "Deep-Resilience" framework was fundamentally enabled by recent breakthroughs in Multimodal Foundation Models. Prior attempts to quantify cognitive reserve relied heavily on linear regression models using static proxies like education or IQ. These statistical approaches fail to capture the complex, non-linear interactions between high-dimensional neuroimaging features and multifaceted clinical demographics. In this study, AI functions not merely as a classifier but as a sophisticated "Interpretation Engine." By integrating the Med3DVLM-based 3D Vision Encoder with the Qwen2.5VL Large Language Model, we projected visual brain pathology and clinical context into a unified latent space. The application of LoRA allowed us to efficiently fine-tune this billion-parameter intelligence, enabling the model to learn subtle, non-linear discrepancies that human observers or unimodal models might miss. This study demonstrates that AI is no longer just a tool for automation but a critical enabler for operationalizing abstract neurological theories (like Cognitive Reserve) into computable, clinically actionable metrics.

5.3 Limitations and Strategic Expansion: Towards a Global DPKR Ecosystem

Validation Phase and Demographic Challenges While our initial Phase 1 research demonstrated state-of-the-art performance using the ADNI dataset which chosen for its high accessibility as a global standard benchmark, we recognize the demographic limitations inherent in North American-centric open datasets. Neurodegenerative patterns and genetic risk factors can vary significantly across ethnic groups, potentially limiting the direct applicability of the current model to non-Western populations. Therefore, we position the current results as a successful validation of the Deep-Resilience methodology itself, serving as a foundation for broader application.

Dual-Track Evolution: From Korean-Specific to Global Models To transcend these boundaries and establish a truly race-agnostic cognitive reserve model, we are actively conducting a multi-year research project aimed at a dual-track evolution. First, we plan to fine-tune the model into a 'Korean-specific Cognitive Reserve Model' using high-quality domestic datasets such as the KBASE and SalusCare cohorts. Second, by amalgamating these diverse ethnic datasets via our Privacy-Preserving FL pipeline, we aim to construct a 'Globally Generalized Model' that is robust to diverse genetic and cultural backgrounds without compromising patient privacy.

Integration via DPKR and DPUK Benchmarking Crucially, this strategic expansion is anchored in the development of Dementia Platform Korea (DPKR). Benchmarking the data integration success of Dementia Platform UK (DPUK)³³, we are currently collaborating with DPUK experts (e.g., Dr. Sarah Bauermeister) to standardize and harmonize fragmented clinical data from domestic

hospitals. This initiative allows us to scale our AI-driven cognitive reserve system from a single-dataset pilot into a universal standard for precision dementia care. By creating a collaborative feedback loop between our Multimodal Foundation Models and human clinical experts across these platforms, we will ultimately demonstrate the superior predictive capability of our system in real-world, multi-ethnic clinical settings.

5.4 Societal Impact

As the prevalence of AD accelerates with global aging, the economic and social burden of caregiving is projected to skyrocket³⁴. By identifying individuals with low cognitive reserve prior to the onset of severe symptoms, our AI-driven approach paves the way for preventative interventions (e.g., cognitive training, lifestyle modification). This proactive strategy aligns with the broader societal goal of extending "healthy life expectancy," demonstrating the profound value of integrating advanced AI with geriatric healthcare.

6 Conclusion

In this study, we introduced Deep-Resilience, a multimodal foundation model framework that quantifies cognitive reserve by learning the non-linear discrepancy between brain structure and function. By achieving state-of-the-art performance on the ADNI dataset, we demonstrated that integrating 3D neuroimaging with clinical text via parameter-efficient fine-tuning significantly enhances the prediction of dementia progression, particularly in the heterogeneous MCI stage. Beyond a technical validation, this work lays the groundwork for a paradigm shift from binary diagnosis to personalized prognosis. Moving forward, we aim to evolve this system into a robust, globally applicable solution by integrating multi-ethnic cohorts through the DPKR project and FL pipelines, ultimately contributing to a precision medicine ecosystem that transcends demographic boundaries.

Acknowledgments and Disclosure of Funding

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A Appendix / supplemental material

Code availability (anonymized for review).

To preserve the integrity of double-blind review, we provide an anonymized GitHub repository containing the implementation of the proposed model and inference scripts. The repository is accessible at: <https://anonymous.4open.science/r/Deep-Resilience>

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification:

- We clearly state our contributions in Section 1.2. The proposed architecture and training strategy are detailed in Section 3, empirical validation is provided in Section 4, and the broader scope including the DPKR initiative is discussed in Section 5.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification:

- We explicitly discuss the demographic limitations of using the North American-centric ADNI dataset and the potential lack of generalization to non-Western populations in Section 5.3. We also outline our plan to address this via the DPKR initiative and Federated Learning.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification:

- This paper is an empirical study focusing on the architectural design and experimental validation of the Deep-Resilience framework; it does not present new theoretical theorems or formal proofs.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification:

- We utilize the publicly available ADNI dataset as detailed in Section 4.1. Furthermore, we provide a complete description of the model architecture, prompt templates, and training hyperparameters (e.g., learning rate, batch size, epoch schedules) in Section 3.1 and Section 3.2.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification:

Guidelines:

- We provide the source code, including data preprocessing and training scripts, via an

anonymized GitHub repository linked in the supplementary material. While the ADNI dataset cannot be redistributed directly due to license restrictions, we provide detailed instructions on how to access the data from the official LONI repository and process it using our scripts.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification:

- We detail the data splitting strategy (80% training / 20% validation) in Section 4.1. All training configurations, including hyperparameters (learning rate, batch size), epoch schedules, and loss functions, are specified in Section 3.2.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification:

- Due to the high computational cost of training 3D multimodal foundation models, we report results from a single fixed stratified split. Repeating experiments with multiple seeds to generate error bars was computationally prohibitive.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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- We detail the computing infrastructure, including the specific GPU model (NVIDIA H100 80GB) and CPU specifications (Intel Xeon Platinum 8480+), in Section 3.2.

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Justification:

- We confirm compliance with the NeurIPS Code of Ethics. Our research utilizes de-identified data (ADNI) and adheres to IRB protocols for private cohorts to ensure patient privacy. We also explicitly address potential demographic biases and the limitations of race-agnostic modeling in Section 5.3.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification:

- We discuss positive societal impacts, such as preventative interventions and reduced caregiving burdens, in Section 5.4. We also address potential negative impacts

related to fairness and demographic bias in Section 5.3, proposing a Federated Learning approach to mitigate these risks across diverse populations.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification:

- The proposed model is a specialized medical diagnostic system designed for clinical prognosis. It does not pose the high risks for misuse (e.g., disinformation, harmful content generation) typically associated with general-purpose generative models or uncured scraped datasets.

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Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

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Justification:

- We cite the original creators of all assets used, including the ADNI dataset [29] and the Qwen2.5VL-7B model [27]. We explicitly adhere to the ADNI Data Use Agreement and the respective open-source licenses (e.g., Apache 2.0) for the pre-trained models and codebases.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification:

- We introduce the Deep-Resilience framework as a new asset. The full source code, including documentation for data preprocessing, model training, and inference, is provided in the supplementary material (via an anonymized repository)

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification:

- This study focuses on the analysis of clinical datasets (ADNI, SalusCare) and did not involve crowdsourcing experiments or user studies requiring participant compensation or specific task instructions (e.g., screenshots).

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [Yes]

Justification:

- This study involving human subjects was conducted in accordance with the Declaration of Helsinki and was approved by the Institutional Review Boards (IRB) of the respective participating institutions. All data utilized in this study, including ADNI and the validation cohorts, were de-identified to protect patient privacy.